

Macroeconometrics

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1 Time Trends

1.1 Properties

Definition 1.1 (Time Trend). A time trend is a response variable y_t dependent on t

$$y_t = \alpha + \delta t + u_t$$

where $u_t \sim i.i.d.(0, \sigma^2)$

Proposition 1.2 (Stochastic properties of time trends). Consider the following time trend model

1. Expectation: $E[y_t] = \alpha + \delta t$.
2. Error Term: $y_t - E[y_t] = u_t$
3. Variance: $\text{Var}(y_t) = \sigma^2$
4. Covariance: $\text{Cov}(y_t, y_{t-h}) = 0$ for all $h \in \mathbb{Z}$

1.2 OLS estimation and Asymptotics

We may represent the model in matrix form

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{U}$$

where

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_T \end{pmatrix}, \mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ \vdots & \vdots \\ 1 & T \end{pmatrix}, \beta = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \mathbf{U} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_T \end{pmatrix}$$

where $\mathbf{y}$ is a $T \times 1$ matrix, $\mathbf{X}$ is a $T \times 2$ matrix, β is a 2×1 matrix and $\mathbf{U}$ is a $T \times 1$ matrix.

Proposition 1.3. Consider the OLS estimation problem

$$\hat{\beta}_T = \underset{\beta \in \mathbb{R}^T}{\operatorname{argmin}} \mathbf{U}(\beta)' \mathbf{U}(\beta)$$

$$\text{Then, } \hat{\beta}_T = \begin{pmatrix} \hat{\alpha}_T \\ \hat{\delta}_T \end{pmatrix} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y}$$

Remark. By substituting the model for $\mathbf{y}$, we can write

$$(\hat{\beta}_T - \beta) = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{U}$$

or equivalently

$$\begin{pmatrix} \hat{\alpha}_T - \alpha \\ \hat{\delta}_T - \delta \end{pmatrix} = \begin{pmatrix} \sum_{t=1}^T 1 & \sum_{t=1}^T t \\ \sum_{t=1}^T t & \sum_{t=1}^T t^2 \end{pmatrix}^{-1} \begin{pmatrix} \sum_{t=1}^T u_t \\ \sum_{t=1}^T t u_t \end{pmatrix}$$

To derive the asymptotic distribution, we need to rescale $\hat{\beta}_T$. Let

$$\mathbf{y}_T = \begin{pmatrix} T^{1/2} & 0 \\ 0 & T^{3/2} \end{pmatrix}$$

so that

$$\mathbf{y}_T(\hat{\beta}_T - \beta) = (\mathbf{y}_T^{-1} \mathbf{X}' \mathbf{X} \mathbf{y}_T^{-1})^{-1} \mathbf{y}_T^{-1} \mathbf{X}' \mathbf{U}$$

or equivalently,

$$\begin{aligned} \mathbf{y}_T(\hat{\beta}_T - \beta) &= \begin{pmatrix} T^{1/2}(\hat{\alpha}_T - \alpha) \\ T^{3/2}(\hat{\delta}_T - \delta) \end{pmatrix} \\ &= \begin{pmatrix} T^{-1} \sum_{t=1}^T 1 & T^{-2} \sum_{t=1}^T t \\ T^{-2} \sum_{t=1}^T t & T^{-3} \sum_{t=1}^T t^2 \end{pmatrix}^{-1} \begin{pmatrix} T^{-1/2} \sum_{t=1}^T u_t \\ T^{-3/2} \sum_{t=1}^T t u_t \end{pmatrix} \end{aligned}$$

Notice that

$$\begin{aligned} \mathbf{X}'\mathbf{X} &= \begin{pmatrix} \sum_{t=1}^T 1 & \sum_{t=1}^T t \\ \sum_{t=1}^T t & \sum_{t=1}^T t^2 \end{pmatrix} \\ &= \begin{pmatrix} T & T(T+1)/2 \\ T(T+1)/2 & T(T+1)(2T+1)/6 \end{pmatrix} \end{aligned}$$

Hence,

$$\begin{aligned} \frac{\mathbf{X}'\mathbf{X}}{T^3} &= \frac{1}{T^3} \begin{pmatrix} \sum_{t=1}^T 1 & \sum_{t=1}^T t \\ \sum_{t=1}^T t & \sum_{t=1}^T t^2 \end{pmatrix} \\ &= \frac{1}{T^3} \begin{pmatrix} T & T(T+1)/2 \\ T(T+1)/2 & T(T+1)(2T+1)/6 \end{pmatrix} \end{aligned}$$

so that in the limit when $T \rightarrow \infty$,

$$\frac{\mathbf{X}'\mathbf{X}}{T^3} \longrightarrow \begin{pmatrix} 0 & 0 \\ 0 & 1/3 \end{pmatrix}$$

which is not invertible, which is bad since we initially wanted $(\mathbf{X}'\mathbf{X})^{-1}$.

Instead, we scale such that $\mathbf{y}_T^{-1}(\mathbf{X}'\mathbf{X})\mathbf{y}_T^{-1}$, then

$$\begin{aligned} \mathbf{y}_T^{-1}(\mathbf{X}'\mathbf{X})\mathbf{y}_T^{-1} &= \begin{pmatrix} T^{-1} \sum_{t=1}^T 1 & T^{-2} \sum_{t=1}^T t \\ T^{-2} \sum_{t=1}^T t & T^{-3} \sum_{t=1}^T t^2 \end{pmatrix} \\ &= \begin{pmatrix} T^{-1}T & T^{-2}T(T+1)/2 \\ T^{-2}T(T+1)/2 & T^{-3}T(T+1)(2T+1)/6 \end{pmatrix} \\ &\longrightarrow \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{pmatrix} \equiv \mathbf{Q} \end{aligned}$$

Before we prove asymptotics, we need to know certain facts.

Proposition 1.4. Since $\left(\frac{t}{T}\right)^v$ is a continuous function,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{i=1}^T \left(\frac{t}{T}\right)^v = \int_0^1 r^v dr = \frac{1}{v+1}$$

or equivalently,

$$\frac{1}{T} \sum_{i=1}^T \left(\frac{t}{T}\right)^v \rightarrow \int_0^1 r^v dr = \frac{1}{v+1}$$

Furthermore, notice that the asymptotic distribution of the second term $\mathbf{y}_T^{-1}\mathbf{X}'\mathbf{U}$ cannot be found with just the normal CLT. Notice that in our matrix,

$$\mathbf{y}_T^{-1}\mathbf{X}'\mathbf{U} = \begin{pmatrix} T^{-1/2} \sum_{t=1}^T u_t \\ T^{-3/2} \sum_{t=1}^T t u_t \end{pmatrix} = \begin{pmatrix} (1/\sqrt{T}) \sum_{t=1}^T u_t \\ (1/\sqrt{T}) \sum_{t=1}^T (t/T) u_t \end{pmatrix}$$

the second entry $(1/\sqrt{T}) \sum_{t=1}^T \frac{t}{T} u_t$ does not satisfy the assumption of the Linderberg-Levy CLT since

$$\text{Var}[(t/T)u_t] = \frac{t^2}{T^2} \text{Var}(u_t) = \frac{t^2}{T^2} \sigma^2$$

and so grows as t increases. Hence, we need new asymptotic theory

Definition 1.5 (Filtration). Suppose we have a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ where the event space $\mathcal{F}$ is organised as a σ -field. A filtration $\mathcal{F}_t$ is an increasing sequence of event spaces so that $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$ for $s \leq t$.

Remark. In time series, we denote $\mathcal{F}_t$ as $\mathcal{I}_t$ and call it the information set instead.

Definition 1.6 (Martingale Difference Sequences). Let $\{m_t\}$ be a sequence of random scalars with $E(m_t) = 0$ and let $\mathcal{I}_t$ be the information set available at time t . Then, if

$$E(m_t | \mathcal{I}_{t-1}) = 0$$

then $\{m_t\}$ is said to be a martingale difference sequence with respect to $\{\mathcal{I}_t\}$.

More formally, we say that $(m_t, \mathcal{I}_t)$ is a martingale difference sequence if for all $t \in \mathbb{N}$ then m_t is $\mathcal{I}_t$ -measurable, $E|m_t| < \infty$ and $E(m_t | \mathcal{I}_{t-1}) = 0$

Remark. The intuition is to relax i.i.d. but keep uncorrelatedness and limit the heterogeneity so that some LLN and CLT apply. $E(m_t|\mathcal{I}_{t-1}) = 0$ implies that m_t is serially uncorrelated (m.d.s. assumption is stronger than uncorrelatedness) but weaker than independence since it does not rule out dependence of higher moments/non-linear dependences

Theorem 1.7 (Law of Large Numbers (m.d.s.)). Let $(m_t, \mathcal{I}_t)$ for $t \in \mathbb{N}$ be a martingale difference sequence. If one of the following conditions holds

- (a) $\sum_{t=1}^{\infty} E|m_t|^{1+p}/t^{1+p} < \infty$ for some $p \in [0, 1]$
- (b) $\lim_{T \rightarrow \infty} \max_{t \leq T} E|m_t|^{1+p} < \infty$ for some $p \in [0, 1]$

Then, as $T \rightarrow \infty$

$$T^{-1} \sum_{t=1}^T m_t \xrightarrow{p} 0$$

Theorem 1.8 (Central Limit Theorem (m.d.s.)). Let $(m_t, \mathcal{I}_t)$ for $t \in \mathbb{N}$ be a martingale difference sequence satisfying $Em_t^2 < \infty$ for all t . Let $S_T^2 = \sum_{t=1}^T Em_t^2$. Suppose

- (i) $\sum_{t=1}^T m_t^2/S_T^2 \xrightarrow{p} 1$
- (ii) $\sum_{t=1}^T E|m_t/S_T|^{2+\delta} \rightarrow 0$ for some $\delta > 0$

then

$$\frac{1}{S_T} \sum_{t=1}^T m_t \xrightarrow{d} \mathcal{N}(0, 1)$$

Theorem 1.9 (Cramer-Wold Device/Theorem). Let $\mathbf{X}_T$ be a $p \times 1$ vector. If

$$\lambda' \mathbf{X}_T \xrightarrow{d} \lambda' \mathbf{X}$$

for all $\lambda \in \mathbb{R}^p$, then

$$\mathbf{X}_T \xrightarrow{d} \mathbf{X}$$

Remark. It is actually if and only if (but the backwards direction is easy to see)

Theorem 1.10 (Asymptotics for Time Trends). Let $y_t = \alpha + \delta t + u_t$ where u_t is i.i.d. $(0, \sigma^2)$ and $E(u_t^4) = \mu_4 < \infty$. Then,

$$\begin{pmatrix} T^{1/2}(\hat{\alpha}_T - \alpha) \\ T^{3/2}(\hat{\delta}_T - \delta) \end{pmatrix} \xrightarrow{d} \mathcal{N} \left\{ (0, 0), \sigma^2 \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{pmatrix}^{-1} \right\}$$

Proof. Recall we may write

$$\mathbf{y}_T(\hat{\beta}_T - \beta) = (\mathbf{y}_T^{-1} \mathbf{X}' \mathbf{X} \mathbf{y}_T^{-1})^{-1} \mathbf{y}_T^{-1} \mathbf{X}' \mathbf{U} = \begin{pmatrix} T^{-1} \sum_{t=1}^T 1 & T^{-2} \sum_{t=1}^T t \\ T^{-2} \sum_{t=1}^T t & T^{-3} \sum_{t=1}^T t^2 \end{pmatrix}^{-1} \begin{pmatrix} T^{-1/2} \sum_{t=1}^T u_t \\ T^{-3/2} \sum_{t=1}^T t u_t \end{pmatrix}$$

Since we scaled $\mathbf{X}' \mathbf{X}$ by $\mathbf{y}_T^{-1}$ on each side, then

$$\begin{aligned} \mathbf{y}_T^{-1} (\mathbf{X}' \mathbf{X}) \mathbf{y}_T^{-1} &= \begin{pmatrix} T^{-1} \sum_{t=1}^T 1 & T^{-2} \sum_{t=1}^T t \\ T^{-2} \sum_{t=1}^T t & T^{-3} \sum_{t=1}^T t^2 \end{pmatrix} \\ &= \begin{pmatrix} T^{-1} T & T^{-2} T(T+1)/2 \\ T^{-2} T(T+1)/2 & T^{-3} T(T+1)(2T+1)/6 \end{pmatrix} \\ &\longrightarrow \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{pmatrix} \equiv \mathbf{Q} \end{aligned}$$

The more important part of the proof is the second term of $\mathbf{y}_T(\hat{\beta}_T - \beta)$ which is $\mathbf{y}_T^{-1}\mathbf{X}'\mathbf{U}$. The main goal is to show that if $u_t \sim \text{i.i.d.}(0, \sigma^2)$ and $E(u_t^4) = \mu_4 < \infty$, then

$$\mathbf{y}_T^{-1}(\mathbf{X}'\mathbf{U}) = \begin{pmatrix} (1/\sqrt{T}) \sum_{t=1}^T u_t \\ (1/\sqrt{T}) \sum_{t=1}^T (t/T)u_t \end{pmatrix} \xrightarrow{d} \mathcal{N}(0, \sigma^2 \mathbf{Q})$$

The first entry of the matrix is quite easy. Since by assumption, $u_t \sim \text{i.i.d.}(0, \sigma^2)$ and $E(u_t^4) = \mu_4 < \infty$, then by the Linderberg-Levy CLT,

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T u_t \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

The focus then is on $(1/\sqrt{T}) \sum_{t=1}^T \frac{t}{T} u_t$ which does not satisfy the assumptions of the Linderberg-Levy CLT since $\frac{t}{T} u_t$ in particular is not i.i.d. with constant variance. There are three steps in the proof

1. Check that $\frac{t}{T} u_t$ is a m.d.s.

This is easy since in the sequence $\frac{t}{T} u_t$, $E[\frac{t}{T} u_t] = 0$. Furthermore,

$$E[(t/T)u_t | \mathcal{I}_{t-1}] = \frac{t}{T} E[u_t | \mathcal{I}_{t-1}] = 0$$

Now, we can define $m_t = (t/T)u_t$.

2. Check that

$$\sum_{t=1}^T \{(t/T)u_t\}^2 / S_T^2 \xrightarrow{p} 1$$

holds, where $S_T^2 = \sum_{t=1}^T E\{[(t/T)u_t]^2\}$.

This is equivalent to checking

$$T^{-1} \sum_{t=1}^T \{(t/T)u_t\}^2 / (S_T^2/T) \xrightarrow{p} 1$$

which is equivalent to showing

$$[T^{-1} \sum_{t=1}^T \{(t/T)u_t\}^2 - (S_T^2/T)] \xrightarrow{p} 0$$

Recall

$$\frac{S_T^2}{T} = \frac{1}{T} \sum_{t=1}^T \frac{t^2}{T^2} E(u_t^2)$$

therefore the quantity of interest is

$$[T^{-1} \sum_{t=1}^T \{(t/T)u_t\}^2 - T^{-1} \sum_{t=1}^T \{(t^2/T^2)E(u_t^2)\}] = \frac{1}{T} \sum_{t=1}^T \frac{t^2}{T^2} \{u_t^2 - E(u_t^2)\}$$

We want to show that

$$\frac{1}{T} \sum_{t=1}^T \frac{t^2}{T^2} \{u_t^2 - E(u_t^2)\} \xrightarrow{p} 0$$

Two possible ways to prove this

- (a) m.d.s. LLN: check conditions hold for

$$m_t = \frac{t^2}{T^2} \{u_t^2 - E(u_t^2)\}$$

Step 1: Check if m_t thus defined is a martingale difference sequence. So, wts $E(m_t|\mathcal{I}_{t-1}) = 0$. So,

$$\begin{aligned} E(m_t|\mathcal{I}_{t-1}) &= E[(t^2/T^2)\{u_t^2 - E(u_t^2)\}|\mathcal{I}_{t-1}] \\ &= (t^2/T^2)E[\{u_t^2 - E(u_t^2)\}|\mathcal{I}_{t-1}] \\ &= 0 \end{aligned}$$

by assumption, since $E[u_t^2|\mathcal{I}_{t-1}] = E[u_t^2] = \sigma^2$.

Step 2: check condition (a) i.e. choose $p = 1$ s.t.

$$\begin{aligned} \sum_{t=1}^{\infty} E(m_t^2)/t^2 &= \sum_{t=1}^{\infty} E[(t^2/T^2)^2\{u_t^2 - E(u_t^2)\}^2]/t^2 \\ &= \sum_{t=1}^{\infty} E[\{u_t^2 - E(u_t^2)\}^2](t^2/T^4) \end{aligned}$$

Note that $E(u_t^4) = \mu_4 < \infty$, hence $E[\{u_t^2 - E(u_t^2)\}^2]$ must be a constant, say $E[\{u_t^2 - E(u_t^2)\}^2] = C$. So,

$$\sum_{t=1}^{\infty} E(m_t^2)/t^2 = C \sum_{t=1}^{\infty} (t^2/T^4) = \frac{C}{T^2} \sum_{t=1}^{\infty} (t^2/T^2) \longrightarrow 0$$

Alternative Step 2: check condition (b) i.e. choose $p = 1$ so that

$$\begin{aligned} \lim_{T \rightarrow \infty} \max_{t \leq T} E(m_t)^2 &= \lim_{T \rightarrow \infty} \max_{t \leq T} E[(t^2/T^2)^2\{u_t^2 - E(u_t^2)\}^2] \\ &= \lim_{T \rightarrow \infty} \max_{t \leq T} (t^2/T^2)^2 E[\{u_t^2 - E(u_t^2)\}^2] \end{aligned}$$

Recall again that $E[\{u_t^2 - E(u_t^2)\}^2] = C$ does not depend on t . Hence,

$$\lim_{T \rightarrow \infty} \max_{t \leq T} E(m_t)^2 = C \lim_{T \rightarrow \infty} \max_{t \leq T} (t^2/T^2)^2 = C < \infty$$

Step 3: Apply the LLN to show that

$$T^{-1} \sum_{t=1}^T m_t = \frac{1}{T} \sum_{t=1}^T \frac{t^2}{T^2} \{u_t^2 - E(u_t^2)\} \xrightarrow{p} 0$$

(b) Use a mean square argument to check

$$\text{Var} \left[\frac{1}{T} \sum_{t=1}^T \frac{t^2}{T^2} \{u_t^2 - E(u_t^2)\} \right] \longrightarrow 0$$

This is equivalent to checking

$$D_T = E \left[\frac{1}{T} \sum_{t=1}^T \frac{t^2}{T^2} \{u_t^2 - E(u_t^2)\} \right]^2 \longrightarrow 0$$

The quantity of interest is

$$\begin{aligned} D_T &= \frac{1}{T^2} E \left[\sum_{t=1}^T \frac{t^2}{T^2} \{u_t^2 - E(u_t^2)\} \right]^2 \\ &= \frac{1}{T^2} E \left[\sum_{t=1}^T \sum_{s=1}^T \frac{t^2}{T^2} \{u_t^2 - E(u_t^2)\} \frac{s^2}{T^2} \{u_s^2 - E(u_s^2)\} \right] \\ &= \frac{1}{T^2} \sum_{t=1}^T \sum_{s=1}^T \frac{t^2}{T^2} \frac{s^2}{T^2} E[\{u_t^2 - E(u_t^2)\} \{u_s^2 - E(u_s^2)\}] \end{aligned}$$

Since $u_i \sim \text{i.i.d.}(0, \sigma^2)$ with $E(u_t^4) = \mu_4$, therefore

$$\begin{aligned} D_T &= \frac{1}{T^2} \sum_{t=1}^T \sum_{s=1}^T \frac{t^2}{T^2} \frac{s^2}{T^2} E[\{u_t^2 - E(u_t^2)\}\{u_s^2 - E(u_s^2)\}] \\ &= \frac{1}{T^2} \sum_{t=1}^T \frac{t^4}{T^4} E[\{u_t^2 - E(u_t^2)\}^2] \\ &= E[\{u_t^2 - E(u_t^2)\}^2] \frac{1}{T^2} \sum_{t=1}^T \frac{t^4}{T^4} \end{aligned}$$

Recall that

$$\frac{1}{T} \sum_{t=1}^T \frac{t^4}{T^4} \rightarrow \frac{1}{5}$$

therefore

$$D_T = E[\{u_t^2 - E(u_t^2)\}^2] \frac{1}{T^2} \sum_{t=1}^T \frac{t^4}{T^4} \rightarrow 0$$

Hence, this verifies

$$\sum_{t=1}^T \{(t/T)u_t\}^2 / S_T^2 \xrightarrow{p} 1$$

3. Check that

$$\sum_{t=1}^T E|(t/T)u_t / S_T|^{2+\delta} \rightarrow 0$$

holds, where $S_T^2 = \sum_{t=1}^T E\{[(t/T)u_t]^2\}$.

Choose $\delta = 2$, so that we need to show that

$$\sum_{t=1}^T \frac{(t/T)^4 E(u_t^4)}{S_T^4} \rightarrow 0$$

We assumed that $E(u_t^4) = \mu_4$, hence,

$$\sum_{t=1}^T \frac{(t/T)^4 E(u_t^4)}{S_T^4} = \frac{\mu_4}{S_T^4} \frac{1}{T} \sum_{t=1}^T \left(\frac{t}{T}\right)^4 = \frac{\mu_4}{S_T^4/T} \frac{1}{T} \frac{1}{t=1} \left(\frac{t}{T}\right)^4$$

We have that

$$\frac{1}{T} \frac{1}{t=1} \left(\frac{t}{T}\right)^4 \rightarrow \int_0^1 r^4 dr = \frac{1}{4+1} = \frac{1}{5}$$

and also

$$S_T^4 \frac{1}{T} = \left\{ \frac{(\sigma^2)^2 (T+1)^2 (2T+1)^2}{6^2 T^2} \right\} \frac{1}{T} \rightarrow \infty$$

In combination,

$$\sum_{t=1}^T \frac{(t/T) E(u_t^4)}{S_T^4} = \frac{\mu_4}{S_T^4/T} \frac{1}{T} \sum_{t=1}^T \left(\frac{t}{T}\right)^4 \rightarrow 0$$

Verified.

4. We apply the CLT to show that for $m_t = (t/T)u_t$,

$$\frac{1}{S_T} \sum_{t=1}^T m_t \xrightarrow{d} \mathcal{N}(0, 1)$$

where $S_T^2 = \sum_{t=1}^T \mathbb{E} m_t^2$. Multiplying and dividing by $\sqrt{T}$ yields

$$\frac{1}{S_T/\sqrt{T}} \frac{1}{\sqrt{T}} \sum_{t=1}^T m_t \xrightarrow{d} \mathcal{N}(0, 1)$$

analysing the terms we know that

$$S_T = \sqrt{\sum_{t=1}^T \mathbb{E} m_t^2}$$

to yield

$$S_T/\sqrt{T} = \sqrt{\frac{1}{T} \sum_{t=1}^T \frac{t^2}{T^2} \mathbb{E}(u_t^2)} = \sigma \sqrt{\frac{1}{T} \sum_{t=1}^T \frac{t^2}{T^2}} \rightarrow \sigma \sqrt{\frac{1}{3}}$$

Then, this implies

$$\frac{1}{S_T/\sqrt{T}} \frac{1}{\sqrt{T}} \sum_{t=1}^T \frac{t}{T} u_t = \sigma \sqrt{\frac{1}{3}} \frac{1}{\sqrt{T}} \sum_{t=1}^T \frac{t}{T} u_t$$

We therefore can write

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T \frac{t}{T} u_t \xrightarrow{d} \mathcal{N}(0, \sigma^2/3)$$

Thus far, for our matrix

$$\mathbf{y}_T^{-1}(\mathbf{X}'\mathbf{U}) = \begin{pmatrix} (1/\sqrt{T}) \sum_{t=1}^T u_t \\ (1/\sqrt{T}) \sum_{t=1}^T (t/T) u_t \end{pmatrix}$$

we have shown that by the Linderberg-Levy CLT,

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T u_t \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

and by the martingale difference sequence CLT

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T \frac{t}{T} u_t \xrightarrow{d} \mathcal{N}(0, \sigma^2/3)$$

We need to consider the joint distribution via the Cramer-Wold Theorem. Let $\lambda = (\lambda_1, \lambda_2)' \in \mathbb{R}^2$. We have

$$\lambda' \mathbf{y}_T^{-1}(\mathbf{X}'\mathbf{U}) = (\lambda_1 \quad \lambda_2) \begin{pmatrix} (1/\sqrt{T}) \sum_{t=1}^T u_t \\ (1/\sqrt{T}) \sum_{t=1}^T (t/T) u_t \end{pmatrix}$$

which may be written as

$$\lambda' \mathbf{y}_T^{-1}(\mathbf{X}'\mathbf{U}) = (1/\sqrt{T}) \sum_{t=1}^T \{\lambda_1 + \lambda_2(t/T)\} u_t$$

Using the m.d.s. CLT on the above, we find that for all $\lambda \in \mathbb{R}^2$,

$$\lambda' \mathbf{y}_T^{-1}(\mathbf{X}'\mathbf{U}) \xrightarrow{d} \lambda' \mathcal{N}(0, \sigma^2 \mathbf{Q})$$

Hence,

$$\mathbf{y}_T^{-1}(\mathbf{X}'\mathbf{U}) \xrightarrow{d} \mathcal{N}(0, \sigma^2 \mathbf{Q})$$

Specifically,

$$\lambda' \mathbf{y}_T^{-1}(\mathbf{X}'\mathbf{U}) = \begin{pmatrix} (1/\sqrt{T}) \sum_{t=1}^T u_t \\ (1/\sqrt{T}) \sum_{t=1}^T (t/T) u_t \end{pmatrix} \xrightarrow{d} \mathcal{N}(0, \sigma^2 \mathbf{Q})$$

where remember

$$\mathbf{Q} = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{pmatrix}$$

To see why the covariance in $\mathbf{Q}$ is 1/2 notice that

$$\lim_{T \rightarrow \infty} \text{Cov} \left\{ (1/\sqrt{T}) \sum_{t=1}^T u_t, (1/\sqrt{T}) \sum_{t=1}^T (t/T) u_t \right\} = \frac{1}{T} \text{E} \left\{ \sum_{t=1}^T u_t \sum_{t=1}^T (t/T) u_t \right\}$$

Then,

$$\begin{aligned} \frac{1}{T} \text{E} \left\{ \sum_{t=1}^T u_t \sum_{t=1}^T (t/T) u_t \right\} &= \frac{1}{T} \text{E} \left\{ \sum_{t=1}^T \sum_{s=1}^T u_t (s/T) u_s \right\} \\ &= \frac{1}{T} \text{E} \left\{ \sum_{t=1}^T (t/T) u_t^2 \right\} \\ &= \frac{1}{T} \sum_{t=1}^T (t/T) \text{E}(u_t^2) \\ &= \sigma^2 \frac{1}{T} \sum_{t=1}^T (t/T) \rightarrow \sigma^2 \frac{1}{2} \end{aligned}$$

Taking it all together,

$$\mathbf{y}_T(\hat{\beta}_T - \beta) = (\mathbf{y}_T^{-1} \mathbf{X}' \mathbf{X} \mathbf{y}_T^{-1})^{-1} \mathbf{y}_T^{-1} (\mathbf{X}' \mathbf{U})$$

Therefore,

$$\begin{pmatrix} T^{1/2}(\hat{\alpha}_T - \alpha) \\ T^{3/2}(\hat{\delta}_T - \delta) \end{pmatrix} \xrightarrow{d} \mathcal{N}\{0, (\mathbf{Q}^{-1} \sigma^2 \mathbf{Q} \mathbf{Q}^{-1})\} = \mathcal{N}(0, \sigma^2 \mathbf{Q}^{-1})$$

■

Example 1.11 (With Normal Errors). Assume now that $u_t \sim \mathcal{N}(0, \sigma^2)$. Then, we can appeal to the Cramer-Wold device directly without using m.d.s. LLNs and CLTs.

1.3 Testing

Consider the model

$$y_t = \alpha + \delta t + u_t$$

Our relevant tests are

1. **Constant:** α

- Our Null Hypothesis is

$$H_0 : \alpha = \alpha_0$$

- Test statistic

$$t_\alpha = \frac{\hat{\alpha}_T - \alpha_0}{\left\{ s_T^2 \begin{pmatrix} 1 & 1 \end{pmatrix} (\mathbf{X}' \mathbf{X})^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}^{1/2}}$$

where

$$s_T^2 = \frac{1}{T_2} \sum_{t=1}^T (y_t - \hat{\alpha}_T - \hat{\delta}_T t)^2$$

- Asymptotic Distribution

$$t_\alpha = \frac{\hat{\alpha}_T - \alpha_0}{\left\{ s_T^2 \begin{pmatrix} 1 & 1 \end{pmatrix} (\mathbf{X}' \mathbf{X})^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}^{1/2}} \xrightarrow{d} \mathcal{N}(0, 1)$$

2. Slope: δ

- Null Hypothesis:

$$H_0 : \delta = \delta_0$$

- Test statistic:

$$t_\delta = \frac{\hat{\delta}_T - \delta_0}{\left\{ s_T^2 (1 \quad 1) (\mathbf{X}'\mathbf{X})^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}^{1/2}}$$

where again

$$s_T^2 = \frac{1}{T-2} \sum_{t=1}^T (y_t - \hat{\alpha}_T - \hat{\delta}_T t)^2$$

- Asymptotic distribution

$$t_\delta = \frac{\hat{\delta}_T - \delta_0}{\left\{ s_T^2 (1 \quad 1) (\mathbf{X}'\mathbf{X})^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}} \xrightarrow{d} \mathcal{N}(0, 1)$$

Remark. $\hat{\alpha}_T$ and $\hat{\delta}_T$ converge at different rates; the corresponding standard errors incorporate the different orders of T and so the usual OLS t tests are asymptotically valid

2 Stationary AR(1) Model

2.1 Basic Properties

Definition 2.1. We define the AR(1) model as

$$y_t = \alpha + \phi y_{t-1} + u_t$$

i.e. a first order difference equation.

Remark. We may represent the AR(1) model via its “initial value” through backward substitution (just like any first order difference equation). So,

$$\begin{aligned} y_t &= \phi y_{t-1} + u_t \\ &= \phi(\phi y_{t-2} + u_{t-1}) + u_t \\ &= \phi^2(\phi y_{t-3} + u_{t-2}) + \phi u_{t-1} + u_t \\ &\vdots \\ &= \phi^t y_0 + \sum_{j=0}^{t-1} \phi^j u_{t-j} \end{aligned}$$

Definition 2.2 (Stationarity). The stochastic process $\{y_t\}$ is

- weakly stationary if (i) its mean $E[y_t]$, (ii) its variance $\text{Var}(y_t)$ and (iii) its auto-covariance $\text{Cov}(y_t, y_{t-h})$ for $h \in \mathbb{Z}$ do not depend on t
- strongly/strictly stationary if for every $k \geq 0$ and every $s, t \geq 1$, we have

$$(y_t, y_{t+1}, \dots, y_{t+k}) =_d (y_s, y_{s+1}, \dots, y_{s+k})$$

where $=_d$ denotes ‘has the same distribution as’

Furthermore, $\{y_t\}$ and $\{x_t\}$ are jointly strictly stationary if

$$(x_t, y_t, x_{t+1}, y_{t+1}, \dots, x_{t+k}, y_{t+k}) =_d (x_s, y_s, x_{s+1}, y_{s+1}, \dots, x_{s+k}, y_{s+k})$$

for every $k \geq 0$ and $s, t \geq 1$

Proposition 2.3. If $\{y_t\}$ and $\{x_t\}$ are individually stationary, then $\{x_t\}$ and $\{y_t\}$ are jointly stationary

Definition 2.4. The stationary AR(1) process is

$$y_t = \phi y_{t-1} + u_t$$

where $|\phi| < 1$, $u_t \sim \text{i.i.d.}(0, \sigma^2)$, and y_0 is a random variable independent of u_t for all t with

$$\mathbb{E}[y_0] = 0, \quad \text{Var}(y_0) = \sigma^2 / (1 - \phi^2)$$

We can see that the model satisfies weak stationarity

Proposition 2.5. The AR(1) model

$$y_t = \phi y_{t-1} + u_t$$

where $|\phi| < 1$, $u_t \sim \text{i.i.d.}(0, \sigma^2)$, and y_0 is a random variable independent of u_t for all t with

$$\mathbb{E}[y_0] = 0, \quad \text{Var}(y_0) = \sigma^2 / (1 - \phi^2)$$

is weakly stationary with the following moments

$$\begin{aligned} \mathbb{E}(y_t) &= 0 \\ \text{Var}(y_t) &= \frac{\sigma^2}{(1 - \phi^2)} \\ \text{Cov}(y_t, y_{t-h}) &= \phi^h \frac{\sigma^2}{(1 - \phi^2)} \\ \text{Corr}(y_t, y_{t-h}) &= \phi^h \end{aligned}$$

Proof. Mean: Consider

$$\begin{aligned} \mathbb{E}(y_t) &= \mathbb{E}(\phi^t y_0 + \sum_{j=0}^{t-1} \phi^j u_{t-j}) \\ &= \phi^t \mathbb{E}(y_0) + \sum_{j=0}^{t-1} \phi^j \mathbb{E}(u_{t-j}) \\ &= 0 \end{aligned}$$

Variance: Consider

$$\text{Var}(y_t) = \text{Var}(\phi^t y_0 + \sum_{j=0}^{t-1} \phi^j u_{t-j})$$

and since y_0 is independent of u_t for all t

$$\text{Var}(y_t) = \text{Var}(\phi^t y_0) + \text{Var}(\sum_{j=0}^{t-1} \phi^j u_{t-j})$$

Notice that the first term

$$\text{Var}(\phi^t y_0) = \phi^{2t} \text{Var}(y_0) = \phi^{2t} \{\sigma^2 / (1 - \phi^2)\}$$

For the second term, notice that since u_t is i.i.d. $(0, \sigma^2)$

$$\text{Var}\left(\sum_{j=0}^{t-1} \phi^j u_{t-j}\right) = \sum_{j=0}^{t-1} \phi^{2j} \text{E}(u_{t-j}^2) = \sigma^2 \sum_{j=0}^{t-1} \phi^{2j}$$

Hence,

$$\text{Var}(y_t) = \phi^{2t} \{\sigma^2 / (1 - \phi^2)\} + \sigma^2 \sum_{j=0}^{t-1} \phi^{2j}$$

By the properties of GP and the stationarity assumption,

$$\sigma^2 \sum_{j=0}^{t-1} \phi^{2j} = \sigma^2 \frac{(1 - \phi^{2t})}{(1 - \phi^2)}$$

Hence,

$$\begin{aligned} \text{Var}(y_t) &= \phi^{2t} \frac{\sigma^2}{(1 - \phi^2)} + \sigma^2 \frac{(1 - \phi^{2t})}{(1 - \phi^2)} \\ &= \sigma^2 \frac{\phi^{2t} + 1 - \phi^{2t}}{(1 - \phi^2)} \\ &= \frac{\sigma^2}{(1 - \phi^2)} \end{aligned}$$

Covariance: We can write the covariance function as

$$\text{Cov}(y_t, y_{t-h}) = \text{E}[y_t y_{t-h}]$$

since $\text{E}[y_t] = 0$. Then,

$$\text{E}[y_t y_{t-h}] = \text{E}\left\{(\phi^t y_0 + \sum_{j=0}^{t-1} \phi^j u_{t-j})(\phi^{t-h} y_0 + \sum_{j=0}^{t-h-1} \phi^j u_{t-h-j})\right\}$$

Since y_0 is independent of u_t for all t ,

$$\begin{aligned} \text{Cov}(y_t, y_{t-h}) &= \text{E}(\phi^{2t-h} y_0^2 + \sum_{j=0}^{t-1} \phi^j u_{t-j} \sum_{j=0}^{t-h-1} \phi^j u_{t-h-j}) \\ &= \phi^{2t-h} \text{Var}(y_0) + \text{E}\left\{\left(\sum_{j=0}^{t-1} \phi^j u_{t-j}\right)\left(\sum_{j=0}^{t-h-1} \phi^j u_{t-h-j}\right)\right\} \end{aligned}$$

Define $A := \text{E}\left\{\left(\sum_{j=0}^{t-1} \phi^j u_{t-j}\right)\left(\sum_{j=0}^{t-h-1} \phi^j u_{t-h-j}\right)\right\}$. Notice that we may expand A to

$$\begin{aligned} A &= \text{E}\left\{\left(\sum_{j=0}^{t-1} \phi^j u_{t-j}\right)\left(\sum_{j=0}^{t-h-1} \phi^j u_{t-h-j}\right)\right\} \\ &= \text{E}\left(\sum_{j=0}^{t-1} \sum_{i=0}^{t-h-1} \phi^h \phi^i u_{t-j} u_{t-h-i}\right) \end{aligned}$$

By linearity of expectations,

$$A = \sum_{j=0}^{t-1} \sum_{i=0}^{t-h-1} \phi^h \phi^i \text{E}(u_{t-j} u_{t-h-i})$$

Notice that by independence, when $t-j \neq t-h-i$, $E(u_{t-j}u_{t-h-i}) = 0$, so when $j = h+1$, $E(u_{t-j}u_{t-h-i}) \neq 0$. We may write

$$\begin{aligned} A &= \sum_{i=0}^{t-h-1} \phi^{h+i} \phi^i \sigma^2 = \sum_{i=0}^{t-h-1} \phi^{h+2i} \sigma^2 = \phi^h \sigma^2 \sum_{i=0}^{t-h-1} \phi^{2i} \\ &= \phi^h \sigma^2 \frac{1 - \phi^{2(t-h)}}{(1 - \phi^2)} \end{aligned}$$

So,

$$\text{Cov}(y_t, y_{t-h}) = \phi^{2t-h} \text{Var}(y_0) + \phi^h \sigma^2 \frac{1 - \phi^{2(t-h)}}{(1 - \phi^2)}$$

Since $\text{Var}(y_0) = \sigma^2 / (1 - \phi^2)$,

$$\begin{aligned} \text{Cov}(y_t, y_{t-h}) &= \phi^{2t-h} \text{Var}(y_0) + \phi^h \{1 - \phi^{2(t-h)}\} \text{Var}(y_0) \\ &= \text{Var}(y_0) (\phi^{2t-h} + \phi^h - \phi^{2t-2h} \phi^h) \\ &= \phi^h \text{Var}(y_0) \end{aligned}$$

It also follows that

$$\text{Corr}(y_t, y_{t-h}) = \frac{\text{Cov}(y_t, y_{t-h})}{\text{Var}(y_t)} = \phi^h$$

■

Definition 2.6 (Lag Polynomials). An AR(1) model

$$y_t = \phi y_{t-1} + u_t$$

can be written using the lag operator as follows

$$(1 - \phi L)y_t = u_t$$

where L is the lag operator. We then denote

$$\Phi(L) = (1 - \phi L)$$

as the lag polynomial.

Hence, we can rewrite the AR(1) as

$$\Phi(L)y_t = u_t$$

Remark. The syntactic grammar of L seems to be like E where

$$L(u_t) = L(u_{t-1})$$

and analogously,

$$L^j(u_t) = L(u_{t-j})$$

The reason for using the lag polynomial representation is because we can see that the nice properties that hold for AR(1) under stationarity can be given another interpretation.

If we invert the polynomial, we get

$$y_t = \{\Phi(L)\}^{-1} u_t = \frac{1}{1 - \phi L} u_t$$

where

$$\frac{1}{1 - \phi L} = 1 + \phi L + \phi^2 L^2 + \dots$$

The nice properties hold for the AR(1) if the roots of the polynomial $\Phi(L) = 1 - \phi L$ are outside the unit circle. The root of the polynomial $1 - \phi z = 0$ is at $z = 1/\phi$. Then the root is outside the unit circle ($|z| > 1$) iff $|\phi| < 1$. If the roots of the polynomial $\Phi(L)$ are outside the unit circle, we can invert the polynomial so that

$$\{\Phi(z)\}^{-1} = \sum_{j=0}^{\infty} \phi^j L^j$$

and $\sum_{j=0}^{\infty} |\phi^j| < \infty$ (this ensures that y_t is a well-defined random variable – the reason is beyond the scope of this course). Applying this infinite series to an infinite series of innovation shocks u_t gives us

$$y_t = \{\Phi(L)\}^{-1} u_t = \sum_{j=0}^{\infty} \phi^j L^j u_t = \sum_{j=0}^{\infty} \phi^j u_{t-j}$$

Hence, there is a Wold Decomposition/MA representation for a stationary AR(1) process (the next proposition).

Proposition 2.7 (Wold Decomposition). Consider a stationary AR(1) process

$$y_t = \phi y_{t-1} + u_t$$

with u_t i.i.d. $(0, \sigma^2)$. Then, there is an equivalent MA representation, called the Wold decomposition, written as

$$y_t = \sum_{j=0}^{\infty} \phi^j u_{t-j}$$

2.2 OLS estimation and Asymptotics

The key insight is that the stationary AR(1) process is not i.i.d. because it is serially correlated i.e. $\text{Cov}(y_t, y_{t-h}) = \phi^h(\sigma^2/(1 - \phi^2)) \neq 0$. So, we need Limit Theorems that loosen the i.i.d. restriction to allow for dependence i.e. have Limit Theorems that only place restrictions on stationarity.

Theorem 2.8 (Law of Large Numbers for Stationary Processes). Let y_t be a covariance-stationary process with $E(y_t) = \mu$ and $\gamma_h = \text{Cov}(y_t, y_{t-h})$ and absolutely summable autocovariances so that $\sum_{h=0}^{\infty} |\gamma_h| < \infty$. Then,

$$\frac{1}{T} \sum_{t=1}^T y_t \xrightarrow{p} \mu$$

as $T \rightarrow \infty$

Example 2.9 (AR(1)). Consider our stationary AR(1) model, $y_t = \phi y_{t-1} + u_t$. Then, our autocovariance function is

$$\gamma_h = \phi^{|h|} \frac{\sigma^2}{(1 - \phi^2)}$$

for all h . We only need to check that γ_h is absolutely summable i.e. $\sum_{h=0}^{\infty} |\gamma_h| < \infty$. So, consider the infinite sum

$$\begin{aligned} \sum_{h=0}^{\infty} |\gamma_h| &= \sum_{h=0}^{\infty} |\phi|^{|h|} \frac{\sigma^2}{|1 - \phi^2|} \\ &= \frac{\sigma^2}{|1 - \phi^2|} \sum_{j=0}^{\infty} |\phi|^j \\ &= \frac{\sigma^2}{|1 - \phi^2|} (1 + |\phi| + |\phi|^2 + |\phi|^3 + \dots) \\ &= \frac{\sigma^2}{|1 - \phi^2|} \frac{1}{1 - |\phi|} < \infty \end{aligned}$$

Hence, the LLN applies for the AR(1) model and

$$\frac{1}{T} \sum_{t=1}^T y_t \xrightarrow{p} 0$$

Theorem 2.10 (Central Limit Theorem). If $y_t = \mu + \Psi(L)u_t$ so that

$$y_t = \mu + \sum_{j=0}^{\infty} \psi_j u_{t-j}$$

where $u_t \sim \text{i.i.d.}(0, \sigma^2)$ and $\sum_{j=0}^{\infty} |\psi_j| < \infty$, then as $T \rightarrow \infty$,

$$\sqrt{T}(\bar{y}_T - \mu) \xrightarrow{d} \mathcal{N}\left(0, \sum_{h=-\infty}^{\infty} \gamma_h\right)$$

where $\sum_{h=-\infty}^{\infty} \gamma_h = \sigma^2 \Phi^2(1)$ is the long run variance

Remark. See Hamilton's Time Series Analysis p.195. Also, this is where the Wold decomposition becomes useful.

Example 2.11 (AR(1)). Here we focus on the stationary AR(1) model $y_t = \phi y_{t-1} + u_t$

Step 1: Check if we can write the model in terms of $y_t = \mu + \Psi(L)u_t$. We can do this for the stationary AR(1) model by the Wold decomposition since

$$y_t = \{\Phi(L)\}^{-1} u_t = \sum_{j=0}^{\infty} \phi^j u_{t-j}$$

Step 2: Check that $\sum_{j=0}^{\infty} |\psi_j| < \infty$ is satisfied. So,

$$\sum_{j=0}^{\infty} |\phi|^j = 1 + |\phi| + |\phi|^2 + \dots = \frac{1}{1 - |\phi|} < \infty$$

Step 3: Figure out the long-run variance. Since $\Psi(L) = \{\Phi(L)\}^{-1}$, then $\Psi(L) = \frac{1}{1-\phi L}$. The long run variance is then

$$\sum_{j=-\infty}^{\infty} \gamma_j = \sigma^2 \Psi^2(1) = \sigma^2 \left(\frac{1}{1 - \phi^2} \right)^2$$

Step 4: Apply the CLT to show that

$$\sqrt{T} \bar{y}_T \xrightarrow{d} \mathcal{N}\left\{0, \frac{\sigma^2}{(1 - \phi)^2}\right\}$$

This allows us to conduct inferences on $E(y_t)$ and do hypothesis testing, confidence intervals etc.

Proposition 2.12. Consider AR(1) model

$$y_t = \phi y_{t-1} + u_t$$

where $u_t \sim \text{i.i.d.}(0, \sigma^2)$ and the corresponding OLS problem

$$\hat{\phi}_T = \underset{\phi \in \mathbb{R}}{\operatorname{argmin}} \sum_{t=1}^T (y_t - \phi y_{t-1})^2$$

The OLS estimator is

$$\hat{\phi}_T = \frac{\sum_{t=1}^T y_t y_{t-1}}{\sum_{t=1}^T y_{t-1}^2}$$

Proof. ■

Remark. We may write the estimator as

$$\hat{\phi} = \phi + \frac{\sum_{t=1}^T y_{t-1} u_t}{\sum_{t=1}^T y_{t-1}^2}$$

Theorem 2.13. Consider the stationary AR(1) model

$$y_t = \phi y_{t-1} + u_t$$

where $u_t \sim \text{i.i.d.}(0, \sigma^2)$. Then,

- Consistency:

$$\hat{\phi}_T \xrightarrow{p} \phi$$

- Asymptotic Normality:

$$\sqrt{T}(\hat{\phi}_T - \phi) \xrightarrow{d} \mathcal{N}\{0, (1 - \phi^2)\}$$

Proof. Consistency: We can divide both the numerator and denominator by T

$$\hat{\phi}_T - \phi = \frac{T^{-1} \sum_{t=1}^T y_{t-1} u_t}{T^{-1} \sum_{t=1}^T y_{t-1}^2}$$

Considering the numerator

$$T^{-1} \sum_{t=1}^T y_{t-1} u_t$$

which is a martingale difference sequence since

$$E(y_{t-1} u_t | \mathcal{I}_{t-1}) = y_{t-1} E(u_t | \mathcal{I}_{t-1}) = 0$$

so that $m_t = y_{t-1} u_t$ is a martingale difference sequence. We check condition (b)

$$\lim_{T \rightarrow \infty} \max_{t \leq T} E|m_t|^{1+p} < \infty$$

for some $p \in [0, 1]$ with $m_t = y_{t-1} u_t$ with $p = 1$. We need to show

$$\lim_{T \rightarrow \infty} \max_{t \leq T} E|y_{t-1} u_t|^2 = \lim_{T \rightarrow \infty} \max_{t \leq T} E(y_{t-1}^2 u_t^2) < \infty$$

By LIE,

$$\begin{aligned} E(y_{t-1}^2 u_t^2) &= E\{E(y_{t-1}^2 u_t^2 | \mathcal{I}_{t-1})\} \\ &= E\{y_{t-1}^2 E(u_t^2 | \mathcal{I}_{t-1})\} \\ &= \sigma^2 E(y_{t-1}^2) \\ &= \sigma^2 \frac{\sigma^2}{1 - \phi^2} \end{aligned}$$

Therefore,

$$\lim_{T \rightarrow \infty} \max_{t \leq T} E(y_{t-1}^2 u_t^2) = \frac{\sigma^4}{1 - \phi^2} < \infty$$

so the m.d.s. LLN applies so that

$$T^{-1} \sum_{t=1}^T y_{t-1} u_t \xrightarrow{p} 0$$

Now consider the denominator

$$T^{-1} \sum_{t=1}^T y_{t-1}^2 = T^{-1} (y_0^2 + y_1^2 + \cdots + y_{T-2}^2 + y_{T-1}^2)$$

Add and subtract y_T^2 so that

$$\begin{aligned} T^{-1} \sum_{t=1}^T y_{t-1}^2 &= T^{-1} (y_0^2 + y_1^2 + \cdots + y_{T-1}^2 + y_T^2 - y_T^2) \\ &= T^{-1} \sum_{t=1}^T y_t^2 + T^{-1} (y_0^2 - y_T^2) \end{aligned}$$

By definition of the AR(1) model

$$y_t^2 = \phi^2 y_{t-1}^2 + u_t^2 + 2\phi y_{t-1} u_t$$

Substituting in

$$\begin{aligned} T^{-1} \sum_{t=1}^T y_{t-1}^2 &= T^{-1} \sum_{t=1}^T y_t^2 + T^{-1} (y_0^2 - y_T^2) \\ &= T^{-1} \sum_{t=1}^T (\phi^2 y_{t-1}^2 + u_t^2 + 2\phi y_{t-1} u_t) + T^{-1} (y_0^2 - y_T^2) \end{aligned}$$

We can then write

$$(1 - \phi^2) T^{-1} \sum_{t=1}^T y_{t-1}^2 = T^{-1} \sum_{t=1}^T u_t^2 + 2\phi T^{-1} \sum_{t=1}^T y_{t-1} u_t + T^{-1} (y_0^2 - y_T^2)$$

Let

$$D_T = T^{-1} \sum_{t=1}^T y_{t-1}^2$$

then

$$D_T = \frac{1}{(1 - \phi^2)} \left\{ T^{-1} \sum_{t=1}^T u_t^2 + 2\phi T^{-1} \sum_{t=1}^T y_{t-1} u_t + T^{-1} (y_0^2 - y_T^2) \right\}$$

Analyse each term inside the curly bracket separately

$$\begin{aligned} T^{-1} \sum_{t=1}^T u_t^2 &\xrightarrow{p} \sigma^2, \quad (\text{i.i.d. LLN}) \\ T^{-1} \sum_{t=1}^T y_{t-1} u_t &\xrightarrow{p} 0, \quad (\text{mds LLN}) \\ T^{-1} (y_0^2 - y_T^2) &\xrightarrow{p} 0, \quad (\text{left as an exercise}) \end{aligned}$$

Therefore, we get

$$T^{-1} \sum_{t=1}^T y_{t-1}^2 \xrightarrow{p} \frac{\sigma^2}{(1 - \phi^2)}$$

Combining we get

$$\hat{\phi}_T - \phi = \frac{T^{-1} \sum_{t=1}^T y_{t-1} u_t}{T^{-1} \sum_{t=1}^T y_{t-1}^2} \xrightarrow{p} \frac{0}{\sigma^2 / (1 - \phi^2)} = 0$$

In short

$$\hat{\phi}_T \xrightarrow{p} \phi$$

Asymptotic Normality: Rewrite the OLS estimator such that

$$\sqrt{T}(\hat{\phi}_T - \phi) = \frac{T^{-1/2} \sum_{t=1}^T y_{t-1} u_t}{T^{-1} \sum_{t=1}^T y_{t-1}^2}$$

For the denominator, we already establish that

$$T^{-1} \sum_{t=1}^T y_{t-1}^2 \xrightarrow{p} \frac{\sigma^2}{(1 - \phi^2)}$$

The numerator is

$$T^{-1/2} \sum_{t=1}^T y_{t-1} u_t$$

with $m_t = y_{t-1} u_t$ being a martingale difference sequence and so the m.d.s. CLT applies. (Left as an exercise). By the m.d.s. CLT is

$$T^{-1/2} \sum_{t=1}^T y_{t-1} u_t \xrightarrow{d} \mathcal{N} \left\{ 0, \frac{\sigma^4}{(1 - \phi^2)} \right\}$$

Combining

$$\frac{T^{-1/2} \sum_{t=1}^T y_{t-1} u_t}{T^{-1} \sum_{t=1}^T y_{t-1}^2} \xrightarrow{d} \mathcal{N} \left[0, \left\{ \frac{\sigma^2}{(1 - \phi^2)} \right\}^{-1} \frac{\sigma^4}{(1 - \phi^2)} \left\{ \frac{\sigma^2}{(1 - \phi^2)} \right\}^{-1} \right]$$

In short,

$$\sqrt{T}(\hat{\phi}_T - \phi) \xrightarrow{d} \mathcal{N}\{0, (1 - \phi^2)\}$$

■

So we can conduct inferences in the usual way, but we need to estimate the asymptotic variance.

3 Random Walks and Unit Roots

Definition 3.1 (Random Walks). A random walk is

$$y_t = \beta + y_{t-1} + u_t$$

where u_t is i.i.d.

We consider random walks as our main kind of unit root processes.

Remark. Notice that for $y_t = \beta + y_{t-1} + u_t$, we can use the lag operator to write it as

$$(1 - L)y_t = \beta + u_t$$

which implies $\phi(L) = 1 - L$. To find the root, we take $\phi(L) = 0$. This means that the autoregressive lag polynomial has a unitary root

$$1 - L = 0$$

3.1 Properties

Assumption 3.2 (Simplified model). Consider the random walk model

$$y_t = \beta + y_{t-1} + u_t$$

(a) $u_t \sim \text{i.i.d.}(0, \sigma^2)$

(b) $y_0 = 0$

The only real assumption is $y_0 = 0$.

Proposition 3.3. Under Assumption 3.2, we may rewrite $y_t = \beta + y_{t-1} + u_t$ as

$$y_t = \beta t + \sum_{j=1}^t u_j$$

Furthermore, our moments are

- Expectation: $E(y_t) = \beta t$
- Variance: $\text{Var}(y_t) = \sigma^2 t$
- Autocovariance: $\text{Cov}(y_t, y_s) = \sigma^2 \min\{t, s\}$

Proof. For the new representation, We show this recursively

$$\begin{aligned} y_t &= \beta + y_{t-1} + u_t \\ &= \beta + (\beta + y_{t-2} + u_{t-1}) + u_t \\ &= 3\beta + y_{t-1} + y_{t-2} + y_{t-3} + u_{t-2} + u_{t-1} + u_t \\ &\vdots \\ &= \beta t + y_0 + \sum_{j=1}^t u_j \\ &= t\beta + \sum_{j=1}^t u_j \end{aligned}$$

Given the representation as $y_t = \beta t + \sum_{j=1}^t u_j$, we can show

1. Expectation: Simply write

$$E(y_t) = E\left(\beta t + \sum_{j=1}^t u_j\right) = \beta t$$

2. Variance: Write

$$\begin{aligned} \text{Var}(y_t) &= \text{Var}\left(\beta t + \sum_{j=1}^t u_j\right) \\ &= \text{Var}\left(\sum_{j=1}^t u_j\right) \\ &= \sigma^2 t \end{aligned}$$

3. Autocovariance: First write

$$\begin{aligned} \text{Cov}(y_t, y_s) &= E[\{y_t - E(y_t)\}\{y_s - E(y_s)\}] \\ &= E\left\{\left(\sum_{j=1}^t u_j\right)\left(\sum_{i=1}^s u_i\right)\right\} \\ &= \sum_{j=1}^t \sum_{i=1}^s E(u_j u_i) \end{aligned}$$

Notice that $E(u_i u_j) = \sigma^2$ if $i = j$ and 0 otherwise. I.e. $E(u_i u_j) = 0$ if $i \neq j$. Hence, we must take into account whether the range of $[1, t]$ or $[1, s]$ is longer:

If $t < s$, then

$$\text{Cov}(y_t, y_s) = \sum_{j=1}^t \sum_{i=1}^s E(u_j u_i) = \sum_{j=1}^t E(u_j^2) = \sigma^2 t$$

and if $s < t$, then

$$\text{Cov}(y_t, y_s) = \sum_{j=1}^t \sum_{i=1}^s E(u_j u_i) = \sum_{i=1}^s E(u_i^2) = \sigma^2 s$$

Hence, $\text{Cov}(y_t, y_s) = \sigma^2 \min\{t, s\}$

■

Remark. The new representation gives us insight into why it is called a random walk (with drift) – it is a cumulation of random, uncorrelated shocks that move in random directions. Shocks have a permanent effect on the future of the series.

Furthermore, the first difference, $\Delta y_t = \beta + u_t$ is stationary.

Standard inference does not hold, unlike the stationary AR(1) process.

3.2 Asymptotics

Here, we focus on the properties of the sample mean $\bar{y}_T = T^{-1} \sum_{t=1}^T y_t$.

Proposition 3.4. Consider a random walk without drift

$$y_t = y_{t-1} + u_t$$

where $u_t \sim \text{i.i.d.}(0, \sigma^2)$. Then, our moments are

- Expectation: $E(\bar{y}_T) = 0$
- Variance: $\text{Var}(\bar{y}_T) = \frac{\sigma^2}{T^2} \frac{T(T+1)(2T+1)}{6} \simeq \frac{\sigma^2}{3} T$

Proof. First, notice that we may rewrite the random walk as a partial sum process

$$y_t = \sum_{j=1}^t u_j$$

Then,

$$E(\bar{y}_T) = \frac{1}{T} \sum_{t=1}^T E(y_t) = \frac{1}{T} \sum_{t=1}^T \sum_{j=1}^t E(u_j) = 0$$

Also,

$$\begin{aligned} \text{Var}(\bar{y}_T) &= \frac{1}{T^2} \left\{ \sum_{t=1}^T \text{Var}(y_t) + 2 \sum_{t=1}^{T-1} \sum_{s=t+1}^T \text{Cov}(y_t, y_s) \right\} \\ &= \frac{1}{T^2} \left\{ \sum_{t=1}^T \sigma^2 t + 2 \sum_{t=1}^{T-1} \sum_{s=t+1}^T \sigma^2 \min\{t, s\} \right\} \\ &= \frac{\sigma^2}{T^2} \left\{ \sum_{t=1}^T t + 2 \sum_{t=1}^{T-1} \sum_{s=t+1}^T t \right\} \\ &= \frac{\sigma^2}{T^2} \left\{ \sum_{t=1}^T t + 2 \sum_{t=1}^{T-1} t \sum_{s=t+1}^T 1 \right\} \\ &= \frac{\sigma^2}{T^2} \left\{ \sum_{t=1}^T t + 2 \sum_{t=1}^{T-1} t(T-t) \right\} \end{aligned}$$

If $t = T$, then the second term is zero, so

$$\begin{aligned}
\text{Var}(\bar{y}_T) &= \frac{\sigma^2}{T^2} \left\{ \sum_{t=1}^T t + 2 \sum_{t=1}^T t(T-t) \right\} \\
&= \frac{\sigma^2}{T^2} \left\{ \sum_{t=1}^T t + 2T \sum_{t=1}^T t - 2 \sum_{t=1}^T t^2 \right\} \\
&= \frac{\sigma^2}{T^2} \left\{ (1+2T) \sum_{t=1}^T t - 2 \sum_{t=1}^T t^2 \right\} \\
&= \frac{\sigma^2}{T^2} \left\{ (1+2T) \frac{T(T+1)}{2} - 2 \frac{T(T+1)(2T+1)}{6} \right\} \\
&= \frac{\sigma^2}{T^2} \frac{T(T+1)(2T+1)}{6}
\end{aligned}$$

■

Remark. The proof of the variance term is very important, especially the expansion of the sum of the variance. In particular, because the covariance matrix is symmetric, when we sum over $t > s$, this is equal to the sum of $s < t$ i.e.

$$\sum_{s < t} \text{Cov}(y_s, y_t) = \sum_{t > s} \text{Cov}(y_s, y_t)$$

Hence, this is why $\text{Var}(\sum_{t=1}^T y_t) = \sum_{t=1}^T \text{Var}(y_t) + 2 \sum_{t=1}^{T-1} \sum_{s=t+1}^T \text{Cov}(y_t, y_s)$

Notice that

$$\text{Var}(\bar{y}_T) = \frac{\sigma^2}{T^2} \frac{T(T+1)(2T+1)}{6} \simeq \frac{\sigma^2}{3} T \rightarrow \infty$$

so the variance of $\bar{y}_T$ diverges to infinity as $T \rightarrow \infty$.

Remark. When we don't care about the actual limiting value, we can just use approximations to the highest degree polynomial to make an argument for divergence/convergence

The divergence of the sample variance implies that we need to rescale $\bar{y}_T$ to $\bar{y}_T/\sqrt{T}$ so that

$$\lim_{T \rightarrow \infty} \text{Var} \left(\frac{\bar{y}_T}{\sqrt{T}} \right) = \lim_{T \rightarrow \infty} \frac{\text{Var}(\bar{y}_T)}{T} = \frac{\sigma^2}{3}$$

This indicates that

$$\frac{\bar{y}_T}{\sqrt{T}} = T^{-3/2} \sum_{t=1}^T y_t$$

has a finite variance in the limit. It remains to find the asymptotic distribution, where there will be two different conditions: under normality of i.i.d. shocks and under i.i.d.

Motivation for non-standard asymptotics: The reason why we cannot use the standard asymptotic theory is because our object of interest is different. Consider that $\frac{\bar{y}_T}{\sqrt{T}}$ may be written as

$$\frac{\bar{y}_T}{\sqrt{T}} = T^{-3/2} \sum_{t=1}^T y_t = T^{-3/2} \sum_{t=1}^T \sum_{j=1}^t u_j$$

Notice that this is

$$\begin{aligned}
\sum_{t=1}^T \sum_{j=1}^t u_j &= u_1 \\
&+ u_1 + u_2 \\
&+ u_1 + u_2 + u_3 \\
&+ \dots \\
&+ u_1 + u_2 + u_3 + u_4 + \dots + u_T
\end{aligned}$$

which implies

$$\sum_{t=1}^T \sum_{j=1}^t u_j = \sum_{t=1}^T u_j \sum_{t=j}^T 1 = \sum_{j=1}^T u_j (T - j + 1)$$

Compared to the standard i.i.d. case of

$$\frac{1}{T^{1/2}} \sum_{j=1}^T u_j$$

Under Normality:

Theorem 3.5 (Asymptotic Distribution under Normality). Consider a random walk without drift

$$y_t = y_{t-1} + u_t$$

Let $u_t \sim \text{i.i.d.} \mathcal{N}(0, \sigma^2)$. Then,

$$\frac{\bar{y}_T}{\sqrt{T}} = T^{-3/2} \sum_{j=1}^T u_j (T - j + 1) \sim \mathcal{N} \left\{ 0, \frac{\sigma^2}{6} \frac{T(T+1)(2T+1)}{T^3} \right\}$$

As $T \rightarrow \infty$,

$$\frac{\bar{y}_T}{\sqrt{T}} \xrightarrow{d} \mathcal{N}(0, \sigma^2/3)$$

Proof. This is simple to see. Since we can write y_t as a sum of u_j i.e. $y_t = \sum_{j=1}^t u_j$, and since $u_j \sim \text{i.i.d.} \mathcal{N}(0, \sigma^2)$, y_t is also normally distributed. Furthermore, since $\bar{y}_T$ is simply $T^{-1} \sum_{t=1}^T y_t$, it is a linear combination of normally distributed r.v.s and hence itself a normally distributed random variable, with mean $E(\bar{y}_T)$ and variance $\text{Var}(\bar{y}_T)$, which we already know. Hence, in finite samples,

$$\frac{\bar{y}_T}{\sqrt{T}} \sim \mathcal{N} \left\{ 0, \frac{\sigma^2}{T^3} \frac{T(T+1)(2T+1)}{6} \right\}$$

As $T \rightarrow \infty$,

$$\frac{\bar{y}_T}{\sqrt{T}} \xrightarrow{d} \mathcal{N}(0, \sigma^2/3)$$

■

Not under Normality; only i.i.d.

We first need to discuss new Limit Theorems

Definition 3.6 (Brownian Motion). A standard Brownian motion $B(r)$ is a continuous time stochastic process, associating each date $r \in [0, 1]$ with the scalar $B(r)$ such that

- (a) $B(0) = 0$
- (b) For any dates $0 \leq r_1 < r_2 < \dots < r_k \leq 1$, the changes $\{B(r_2) - B(r_1)\}, \{B(r_3) - B(r_2)\}, \dots, \{B(r_k) - B(r_{k-1})\}$ are independent multivariate Gaussian with $\{B(s) - B(r)\} \sim \mathcal{N}(0, s - r)$.

(c) For any given realisation, $B(r)$ is continuous in r with probability 1

Theorem 3.7 (Functional Central Limit Theorem). If u_t is i.i.d. $(0, \sigma^2)$, then

$$X_T(r) = \frac{1}{\sqrt{T}} \sum_{j=1}^{[Tr]} u_j \implies \sigma B(r)$$

where $\implies$ denotes weak convergence (convergence in distribution for real random functions and $B(r)$ is a Brownian motion

Theorem 3.8 (Asymptotic Distribution under i.i.d.). Consider a random walk without drift,

$$y_t = y_{t-1} + u_t$$

where $u_t \sim \text{i.i.d.}(0, \sigma^2)$ but is NOT normally distributed. Then,

$$\frac{\bar{y}_T}{\sqrt{T}} = T^{-3/2} \sum_{t=1}^T y_t \xrightarrow{d} \mathcal{N}(0, \sigma^2/3) = \sigma \int_0^1 B(r) dr$$

where $B(r)$ is a Brownian motion.

Proof. The sketch of the proof is as follows: this expresses the limit of

$$\frac{\bar{y}_T}{\sqrt{T}} = T^{-3/2} \sum_{j=1}^T u_j (T - j + 1) = T^{-1} \sum_{t=1}^T T^{-1/2} \sum_{j=1}^t u_j$$

as the integral of a continuous time stochastic process, where we apply the FCLT to $T^{-1/2} \sum_{j=1}^t u_j$ and the CMT to the sample mean component $T^{-1} \sum_{t=1}^T$. Recall that $X_T(r)$ as a function of r can be written as

$$X_T(r) = \begin{cases} 0 & \text{for } 0 \leq r < 1/T, \\ u_1/\sqrt{T} & \text{for } 1/T \leq r < 2/T, \\ (u_1 + u_2)/\sqrt{T} & \text{for } 2/T \leq r < 3/T, \\ \vdots & \vdots \\ (u_1 + \dots + u_T)/\sqrt{T} & \text{for } r = 1. \end{cases}$$

Then, notice that $X_T(r)$ is really a Riemman sum

$$\frac{1}{T} \sum_{t=1}^T \frac{y_t}{\sqrt{T}} = \frac{1}{T} \sum_{t=1}^T \frac{1}{\sqrt{T}} \sum_{j=1}^t u_j = \int_0^1 X_T(r) dr$$

By the FCLT,

$$X_T(r) \implies \sigma B(r)$$

Since the integral is a continuous functional mapping, by the Continuous mapping theorem,

$$\frac{1}{T} \sum_{t=1}^T \frac{y_t}{\sqrt{T}} = \int_0^1 X_T(r) dr \implies \sigma \int_0^1 B(r) dr$$

■

It is instructive to see the applications to the OLS estimator for AR(1) processes.

Example 3.9. Consider the OLS estimation of the AR(1) process

$$y_t = \rho y_{t-1} + u_t$$

where $u_t \sim \text{i.i.d.}(0, \sigma^2)$ with $E(u_t^4) = \mu_4 < \infty$. The OLS estimate of ρ is given by

$$\hat{\rho} = \frac{\sum_{t=1}^T y_{t-1} y_t}{\sum_{t=1}^T y_{t-1}^2} = \rho + \frac{\sum_{t=1}^T y_{t-1} u_t}{\sum_{t=1}^T y_{t-1}^2}$$

If $|\rho| < 1$, then the LLN and CLT can be applied to obtain the asymptotic distribution of the OLS estimator of ρ where

- m.d.s LLN

$$\frac{1}{T} \sum_{t=1}^T y_{t-1}^2 \xrightarrow{p} E(y_{t-1}^2) = \frac{\sigma^2}{1 - \rho^2}$$

- m.d.s. CLT:

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T y_{t-1} u_t \xrightarrow{d} \mathcal{N}\{0, E(y_{t-1}^2 u_t^2)\} = \mathcal{N}\left(0, \frac{\sigma^4}{1 - \rho^2}\right)$$

Therefore,

$$T^{1/2}(\hat{\rho}_T - \rho) \xrightarrow{d} \mathcal{N}(0, 1 - \rho^2)$$

Notice that

- If $\rho = 1$, the distribution collapses as $T^{1/2}(\hat{\rho}_T - \rho) \xrightarrow{p} 0$ so it is not useful for hypothesis testing
- To obtain a non-degenerate asymptotic distribution: $T(\hat{\rho}_T - \rho)$
- Faster than the stationary case ($T^{1/2}$) but slower than the deterministic trend case ($T^{3/2}$)
- Asymptotic distribution when $\rho = 1$ is non-standard

To see the distribution when $\rho = 1$, note that

$$\hat{\rho} - \rho = \hat{\rho} - 1 = \frac{\sum_{t=1}^T y_{t-1} u_t}{\sum_{t=1}^T y_{t-1}^2}$$

For the numerator, notice that $y_t^2 = y_{t-1}^2 + u_t^2 + 2y_{t-1}u_t$. Hence

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T y_{t-1} u_t &= \frac{1}{2} \left\{ \frac{1}{T} \sum_{t=1}^T (y_t^2 - y_{t-1}^2) - \frac{1}{T} \sum_{t=1}^T u_t^2 \right\} \\ &= \frac{1}{2} \left(\frac{1}{T} y_T^2 - \frac{1}{T} \sum_{t=1}^T u_t^2 \right) \xrightarrow{d} \frac{1}{2} \sigma^2 (\chi_1^2 - 1) \end{aligned}$$

This is because $\frac{y_T^2}{T} = \left(\frac{1}{\sqrt{T}} \sum_{j=1}^T u_j \right)^2 \xrightarrow{d} \sigma^2 \chi_1^2$ since by CLT $\frac{1}{\sqrt{T}} \sum_{j=1}^T u_j \xrightarrow{d} \sigma \mathcal{N}(0, 1)$.

For the denominator, by the FCLT and CMT

$$\frac{1}{T^2} \sum_{t=1}^T y_{t-1}^2 \xrightarrow{d} \sigma^2 \int_0^1 B^2(r) dr$$

Therefore,

$$T(\hat{\rho}_T - 1) = \frac{T^{-1} \sum_{t=1}^T y_{t-1} u_t}{T^{-2} \sum_{t=1}^T y_{t-1}^2} \xrightarrow{d} \frac{\frac{1}{2}(\chi_1^2 - 1)}{\int_0^1 B^2(r) dr}$$

i.e. the Dickey-Fuller distribution. The OLS estimator converges at a rate T which is superconsistent and the asymptotic distribution is non-standard

4 Unit Root Testing

4.1 Testing Null of a Unit Root: DF and ADF tests

Dickey-Fuller Test

1. **Hypothesis:** We run the auxillary regression

$$y_t = \rho y_{t-1} + u_t$$

with

$$H_0 : \rho = 1, \quad H_1 : \rho < 1$$

Equivalently,

$$\Delta y_t = \theta y_{t-1} + u_t$$

where $\theta = \rho - 1$ and hence

$$H_0 : \theta = 0, \quad H_1 : \theta < 0$$

2. **Commonly used specifications:**

- (a) No deterministic components: $\Delta y_t = \theta y_{t-1} + u_t$
- (b) Constant term: $\Delta y_t = \alpha + \theta y_{t-1} + u_t$
- (c) Linear Trend: $\Delta y_t = \alpha + \beta t + \theta y_{t-1} + u_t$

Use one that can explain both H_0 and H_1 i.e. if the data looks trended, then (iii) would be a reasonable specification under both hypotheses; otherwise (ii) is recommended

3. **Assumption:** u_t is taken to be uncorrelated or i.i.d.

Example 4.1. Here we want to see how to derive the DF test. Consider $y_t = y_{t-1} + u_t$ where $u_t = \epsilon_t \sim \text{i.i.d.}(0, \sigma^2)$. Then, the auxillary regression is

$$\Delta y_t = \theta y_{t-1} + u_t$$

where $\theta = \rho - 1$. The OLS estimator of the regression is

$$\hat{\theta}_T = \frac{\sum_{t=1}^T y_{t-1} \Delta y_t}{\sum_{t=1}^T y_{t-1}^2}$$

Note that under the null, $\theta = \rho - 1 = 0$, so $\Delta y_t = u_t = \epsilon_t$. The t -statistic is

$$t_\theta = \frac{\hat{\theta}_T}{\hat{\sigma}_{\hat{\theta}_T}}$$

where

$$\hat{\theta}_T = \frac{\sum_{t=1}^T y_{t-1} \epsilon_t}{\sum_{t=1}^T y_{t-1}^2}$$

$$\hat{\sigma}_{\hat{\theta}_T} = \sqrt{s_T^2 \left(\sum_{t=1}^T y_{t-1}^2 \right)^{-1}}$$

and $s_T^2 = \frac{1}{T-1} \sum_{t=1}^T (y_t - \hat{\rho}_T y_{t-1})^2$. This gives us

$$t_\theta = \frac{\hat{\theta}_T}{\hat{\sigma}_{\hat{\theta}_T}} = \frac{T^{-1} \sum_{t=1}^T y_{t-1} \epsilon_t}{\left(T^{-2} \sum_{t=1}^T y_{t-1}^2 \right)^{1/2} s_T}$$

Recall we have, for $y_t = y_{t-1} + u_t$ where $u_t = \epsilon_t \sim \text{i.i.d.}(0, \sigma^2)$, we have

1. $\frac{1}{T} \sum_{t=1}^T y_{t-1} \epsilon_t \xrightarrow{d} (1/2) \sigma^2 (\chi_1^2 - 1)$
2. $\frac{1}{T^2} \sum_{t=1}^T y_{t-1}^2 \xrightarrow{d} \sigma^2 \int_0^1 B^2(r) dr$
3. $s_T^2 \xrightarrow{p} \sigma^2$

and so,

$$t_\theta \xrightarrow{d} \frac{(1/2)(\chi_1^2 - 1)}{\left(\int_0^1 B(r) dr\right)^{1/2}} =^{\text{def}} \text{DF}_z$$

However, the assumption that $u_t = \varepsilon_t \sim \text{i.i.d.}$ is typically violated in economic applications. If we relax the i.i.d. assumption to have $u_t = \Theta(L)\varepsilon_t$ with $\varepsilon_t \sim \text{i.i.d.}$ and if u_t is autocorrelated, then the distribution of the DF test will change.

the Augmented DF test uses parametric correction to allow for autocorrelation in the error.

Augmented Dickey-Fuller Test

1. **Hypothesis:** We run the auxillary regression

$$\Delta y_t = f(t) + \theta y_{t-1} + \sum_{j=1}^{p-1} \varphi_j \Delta y_{t-j} + \varepsilon_t$$

with

$$H_0 : \theta = 0, \quad H_1 : \theta < 0$$

Under $H_0 : \theta = 0$, the test based on the corresponding t-statistic has the same asymptotic distribution as in the non-autocorrelated case. As before, the presence of deterministic components will affect the asymptotic distribution

2. **Commonly used specifications:** As before, the presence of deterministic components will affect the asymptotic distribution i.e.

(a) No deterministic components: $\Delta y_t = \theta y_{t-1} + \sum_{j=1}^{p-1} \varphi_j \Delta y_{t-j} + \varepsilon_t$

(b) Constant term: $\Delta y_t = \alpha + \theta y_{t-1} + \sum_{j=1}^{p-1} \varphi_j \Delta y_{t-j} + \varepsilon_t$

(c) Linear Trend: $\Delta y_t = \alpha + \beta t + \theta y_{t-1} + \sum_{j=1}^{p-1} \varphi_j \Delta y_{t-j} + \varepsilon_t$

3. **Assumption:** instead of no autocorrelation, we assume that $u_t = \Theta(L) = \varepsilon_t$ with $e_t \sim \text{i.i.d.}$

4.2 Testing Null of stationary: KPSS test

The intuition here is to reverse the null and alternative hypothesis of the DF tests (which has a null of unit root).

1. **Hypothesis:** We want to test

$$H_0 : y_t \sim I(0), \quad H_1 : y_t \sim I(1)$$

We have three null hypotheses

$$y_t = \varepsilon_t$$

$$y_t = \mu + \varepsilon_t$$

$$y_t = \mu + \beta t + \varepsilon_t$$

2. **Test Statistic:** We have

$$KPSS = \frac{T^{-2} \sum_{t=1}^T S_t^2}{s^2}$$

where

$$S_t = \sum_{j=1}^t \hat{\varepsilon}_j$$

with $\hat{\varepsilon}_j$ being the OLS residuals in the corresponding models under the null and s^2 is the estimation of the long run variance of $\hat{\varepsilon}_j$ (short run variance + 2 autocovariances) HAC type

$$s^2 = T^{-1} \sum_{t=1}^T \hat{\varepsilon}_t^2 + 2T^{-1} \sum_{s=1}^{m_T} \left(1 - \frac{s}{m_T + 1}\right) \sum_{t=s+1}^T \hat{\varepsilon}_t \hat{\varepsilon}_{t-s}$$

3. **Distribution:** Under the null, if the errors are $I(0)$, then

$$KPSS \xrightarrow{d} D_{KPSS}$$

where the critical values *by simulations) will be provided.

Under the alternative, the test diverges

$$KPSS = \frac{T^{-2} \sum_{t=1}^T S_t^2}{s^2} \xrightarrow{d} \infty$$

4.3 Breaks and unit roots

Motivation: Structural breaks can be confused with unit root processes. In the presence of breaks (in an otherwise stationary series) DF tests are biased toward the nonrejection of a unit roots (i.e. less likely to reject H_0)

Example 4.2. Consider a simple model

$$y_t = \alpha y_{t-1} + \varepsilon_t + 1_{t < \tau+1}$$

where $\tau + 1$ denotes the break date and $|\alpha| < 1$, then the DF tends towards the nonrejection of the null, $H_0 : \alpha = 1$

Perron's Test for structural breaks:

1. **Hypothesis:** We run the auxillary regression

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + \alpha_2 t + \mu_1 1_{(t > \tau)} + \varepsilon_t$$

with

$$H_0 : \alpha_1 = 1, \quad H_1 : \alpha_1 < 1$$

2. **Test statistic, asymptotics:** t -test on α_1 with a non-standard asymptotic distribution derived by Perron and critical values simulated by Perron

Ender's discussion on Perron's test (pg 227-234):

Let the null and alternative hypotheses be

$$H_0 : y_t = \alpha_0 + y_{t-1} + \mu_1 D_p + \varepsilon_t$$

$$H_1 : y_t = \alpha_0 + \alpha_2 t + \mu_2 D_L + \varepsilon_t$$

where $D_p = 1$ if $t = \tau + 1$ and zero otherwise (pulse dummy); $D_L = 1$ if $t > \tau$ and zero otherwise (level dummy). Under the null, $\{y_t\}$ is a unit root process with a one time jump in the level of the sequence in period $t = \tau + 1$ and under the alternative, it is a trend stationary process with one time jump in the intercept.

Implementation

1. Detrend the data by estimating the model under the alternative and call the residuals $\hat{y}_t$. Hence, each value of $\hat{y}_t$ is the residual from the regression $y_t = \alpha_0 + \alpha_2 t + \mu_2 D_L + \hat{y}_t$.

2. Estimate

$$\hat{y}_t = \alpha_1 \hat{y}_{t-1} + \varepsilon_t$$

Under the null of the unit root, the theoretical value of α_1 is 1. Perron showed that when the residuals are i.i.d., the distribution of α_1 depends on the proportion of observations occurring prior to the break. Denote this proportion by $\lambda = \tau/T$ where T is the total number of observations

3. Perform diagnostic checks to determine if the residuals from Step 2 are serially uncorrelated. If there is serial correlation, use the augmented form of the regression:

$$\hat{y}_t = \alpha_1 \hat{y}_{t-1} + \sum_{i=1}^k \beta_i \Delta \hat{y}_{t-i} + \varepsilon_t$$

4. Calculate the t -statistic for the null $\alpha_1 = 1$. This statistic can be compared to the critical values calculated by Perron

Remark. We can incorporate step 1 into step 3 by estimating

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + a_2 t + \mu_2 D_L + \sum_{i=1}^k \beta_i \Delta y_{t-i} + \varepsilon_t$$

The t -statistic for the null $\alpha_1 = 1$ can be compared to the appropriate critical values calculated by Perron.

Remark. This method is quite general and allows for the following additions (non-exhaustive)

- Test null of permanent change in the drift term vs alternative of a change in slope of trend. Here the hypotheses are

$$H_2 : y_t = \alpha_0 + y_{t-1} + \mu_2 D_L + \varepsilon_t$$

$$H_3 : y_t = \alpha_0 + \alpha_2 t + \mu_3 D_T + \varepsilon_t$$

where $D_L = 1$ if $t > \tau$ and $D_T = t - \tau$ for $t > \tau$ and zero otherwise.

- It is possible to combine the two null hypotheses H_0 and H_2 i.e. a change in both level and drift of a unit root process. The null and alternative are

$$H_4 : y_t = \alpha_0 + y_{t-1} + \mu_1 D_p + \mu_2 D_L + \varepsilon_t$$

$$H_5 : y_t = \alpha_0 + \alpha_2 t + \mu_2 D_L + \mu_3 D_T + \varepsilon_t$$

The procedure, as per the original, entails estimating the regression of the alternative and using the residuals, estimating

$$\hat{y}_t = \alpha_1 \hat{y}_{t-1} + e_t$$

Remark. Lee, Huang and Shin (1997) show that **structural breaks also affect the KPSS test**. They state the following proposition: “stationary tests ignoring the existing break diverge and are biased towards rejecting the null hypothesis in favour of the false alternative unit root hypothesis.

There are generalisations of the KPSS test to accommodate structural breaks when testing the null of stationarity, for instance Buseti and Harvey (2001), Lee and Strazicich (2001) and Kurozumi (2002)

5 Spurious Regression

Three Key Papers to know:

- Yule (1926): “Why do we sometimes get nonsense-correlations between time series?”
Summary: Examined proportion of Church of England marriages to all marriages for the years 1866-1911 inclusive and standardised mortality per 1000 persons for the same years. The correlation coefficient between the time series was 0.9512 despite no physical significance between the two.
Yule found that **two independent cumulated random walks tend to have high correlations** between them despite being independent; the frequency distribution of correlations between samples of independent (non-cumulated) random walks or normal series tend to be normal with mean 0 (i.e. mostly clustered around low correlations)
- Granger and Newbold (1974): “Spurious Regressions in Econometrics”
Summary: They considered

$$\begin{aligned}y_t &= y_{t-1} + v_t \\x_t &= x_{t-1} + w_t\end{aligned}$$

with $y_0 = 0, x_0 = 0$ and $v_t \sim \text{i.i.d.}(0, 1)$ independent of $w_t \sim \text{i.i.d.}(0, 1)$. They ran the least squares regression

$$y_t = \hat{\alpha} + \hat{\beta}x_t + \hat{u}_t$$

and found that while the R^2 was high, and became higher as T increased, it had an extremely low value for the Durbin-Watson statistic, indicating high positive autocorrelation even though y_t and x_t are independent by construction.

- Phillips (1986): “Understanding Spurious Regressions in Econometrics”
Summary: Developed an asymptotic theory for regressions with the spurious regressions of the Granger-Newbold type as a special case. Three key insights:
 1. In the Granger-Newbold regressions of independent random walks, the usual t-ratio significance test does not possess a limiting distribution but diverges as the sample size $T \rightarrow \infty$
 2. The Bias in this test is towards the rejection of no relationship (based on a nominal critical value of 1.96) will increase with T
 3. The Durbin-Watson statistic converges in probability to 0, while the regression R^2 has a non-degenerate limiting distribution as $T \rightarrow \infty$

Proposition 5.1 (Spurious Regression). Let $v_t, w_t \sim \text{i.i.d.}(0, 1)$ for $t = 1, \dots, T$ with v_t, w_t also independent. Define $y_t = \sum_{s=1}^t v_s$ and $x_t = \sum_{s=1}^t w_s$. Then

$$\hat{\beta}_{\text{OLS}} = \frac{\sum_{t=1}^T x_t y_t}{\sum_{t=1}^T x_t^2} = \frac{T^{-2} \sum_{t=1}^T x_t y_t}{T^{-2} \sum_{t=1}^T x_t^2} \xrightarrow{d} \frac{D_{xy}}{D_{xx}}$$

and $|t_{\beta=0}| \xrightarrow{d} \infty$. So, as $T \rightarrow \infty$, we reject the null of $\beta = 0$. This is called spurious regression.

Remark. Notice that for unrelated cross-sectional variables, $t_{\beta=0} \xrightarrow{d} \mathcal{N}(0, 1)$. In this case, the regression reveals the unrelatedness while in the spurious case, the regression hides the unrelatedness

Many macroeconomic time series behave like random walks, using the DF test (Nelson and Plosser, 1982)

6 Cointegration (Static Models)

Definition 6.1 (Integrated Series). A series with no deterministic components which has a stationary representation after differencing d times, is said to be integrated of order d , denoted $x_t \sim I(d)$.

Remark. We mostly consider $I(0)$ and $I(1)$ series, but most results generalise to $d > 1$

It is generally true that if $y_t, x_t \sim I(d)$, then the linear combination

$$z_t = x_t - ay_t$$

will be $I(d)$. But Cointegration is the case where the difference of two $I(1)$ cases can be $I(0)$.

Definition 6.2 (Cointegration). The components of the vector $\mathbf{Y}_t$ are said to be co-integrated of order d, b , denoted $\mathbf{Y}_t \sim \text{CI}(d, b)$ if (i) all components of $\mathbf{Y}_t$ are $I(d)$; (ii) there exists a vector $\alpha \neq 0$ such that $z_t = \alpha' \mathbf{Y}_t \sim I(d - b), b > 0$. The vector α is called the cointegrating vector

Notice that in practice, this means that x_t and y_t need to be $I(1)$ and $z_t = y_t - \theta x_t \sim I(0)$.

Remark. Our interpretation is that $y_t - \theta x_t$ is the equilibrium relationship and so $z_t = y_t - \theta x_t$ is the stationary deviation from the equilibrium, when $z_t \sim I(0)$. Hence, y_t and x_t will not move too far away from one another. The presence of cointegration implies a genuine long-run relationship between non-stationary time series.

Example 6.3 (Common Trend Explanation). One common explanation for why two non-stationary series might be cointegrated is due to a common non-stationary trend. Let $W_t \sim I(1)$ and consider

$$\begin{aligned} y_t &= aW_t + u_{yt} \\ x_t &= W_t + u_{xt} \end{aligned}$$

where $u_{xt}, u_{yt} \sim I(0)$. Then, $y_t \sim I(1)$ and $x_t \sim I(1)$. But,

$$y_t - ax_t = aW_t + u_{yt} - a(W_t + u_{xt}) = u_{yt} - au_{xt} \sim I(0)$$

So, stationarity is achieved by the cancellation of the common trend W_t

Testing for Cointegration in Static Models: Engle-Granger

1. Find OLS residuals $\hat{z}_t$ from static regression

$$y_t = \mu + \beta x_t + z_t, \quad t = 1, \dots, T$$

2. Test for no cointegration (null hypothesis) by testing for a unit root

$$\hat{z}_t = \rho \hat{z}_t + u_t$$

assuming $u_t \sim \text{i.i.d.}(0, \sigma_u^2)$. (Can be augmented with further lags).

3. Consider t -statistic $t_{\rho=1}$ from this regression, referred to as a DF test but the critical values are new (Engle-Granger critical values) which depend on properties of y_t and x_t .

If y_t and x_t are unrelated random walks (i.e. under hypothesis of no cointegration), then the critical values of the Engle-Granger DF test with constant only are

- At 2.5%: -3.12
- At 5%: -2.86
- At 10%: -2.57

For the regression

$$y_t = \mu\beta x_t + z_t$$

in the Engle-Granger cointegration test, note that

- the OLS estimates of μ, β are inefficient (but super-consistent) under cointegration when $z_t \sim I(0)$
- Non-standard asymptotic distributions with nuisance parameters
- There could be more than one cointegrating vector
- Need to specify distribution of x_t . There could be feedback from y_t to x_t
- In general, the t -ratio for β will not have a standard normal distribution (unless x_t is strictly exogenous for y_t)

7 Dynamic Models (ARDL models)

We study one main dynamic model i.e. ARDL models

Definition 7.1 (Autoregressive Distributed Lag Models). The general form of an ARDL($p, q_1, \dots, q_k$) is

$$y_t = \alpha_0 + \alpha_1 t + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=0}^k \sum_{i=0}^{q_j} \theta_{j,i} x_{j,t-i} + \varepsilon_t$$

The simplified form of the ARDL(p, q) model with only one explanatory variable is

$$y_t = \alpha_0 + \alpha_1 t + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=0}^q \theta_j x_{t-j} + \varepsilon_t$$

We almost never include the trend component.

Here are some forms of the ARDL model

- ARDL(p, r)

$$y_t = \mu + \gamma_1 y_{t-1} + \dots + \gamma_p y_{t-p} + \beta_0 x_t + \beta_1 x_{t-1} + \dots + \beta_r x_{t-r} + \varepsilon_t$$

- Partial adjustment: only y dynamics

$$y_t = \mu + \beta_0 x_t + \gamma_1 y_{t-1} + \dots + \gamma_p y_{t-p} + \varepsilon_t$$

- Distributed Lags: no y dynamics

$$y_t = \mu + \beta_0 x_t + \beta_1 x_{t-1} + \dots + \beta_r x_{t-r} + \varepsilon_t$$

- ARDL without contemporaneous x_t

$$y_t = \mu + \gamma_1 y_{t-1} + \dots + \gamma_p y_{t-p} + \beta_1 x_{t-1} + \dots + \beta_r x_{t-r} + \varepsilon_t$$

Definition 7.2 (Error Correction Model). Unfortunately there is no formal definition; the question will usually state but its usually that $E\Delta y_t = E\Delta x_t = 0$ and RHS first term expectation also goes to 0

Theorem 7.3 (Granger Representation Theorem). Every (V)AR can be represented as an (V)ECM

This is only roughly speaking

Example 7.4 (ARDL(1,1)). Consider an ARDL(1,1) model

$$y_t = \mu + \gamma_1 y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + \varepsilon_t$$

This can be given an ECM representation

$$\Delta y_t = (\gamma_1 - 1)(y_{t-1} - \theta x_{t-1} - \tau) + \beta_0 \Delta x_t + \varepsilon_t$$

where $\theta = -(\beta_0 + \beta_1)/(\gamma_1 - 1)$ and $\tau = -\mu/(\gamma_1 - 1)$. To derive this, we do the following

1. Subtract y_{t-1} from both sides so that

$$y_t - y_{t-1} = \mu + \gamma_1 y_{t-1} - y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + \varepsilon_t$$

2. We take the common factor y_{t-1} on RHS to get

$$\Delta y_t = \mu + (\gamma_1 - 1)y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + \varepsilon_t$$

3. Add and subtract $\beta_0 x_{t-1}$ on RHS to yield

$$\Delta y_t = \mu + (\gamma_1 - 1)y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + \beta_0 x_{t-1} - \beta_0 x_{t-1} + \varepsilon_t$$

4. Denote $\Delta x_t = x_t - x_{t-1}$ so that

$$\Delta y_t = \mu + (\gamma_1 - 1)y_{t-1} + \beta_0 \Delta x_t + (\beta_0 + \beta_1)x_{t-1} + \varepsilon_t$$

5. Then the ECM representation follows by taking the common factor $\gamma_1 - 1$ s.t.

$$\Delta y_t = (\gamma_1 - 1) \left(y_{t-1} + \frac{\beta_0 + \beta_1}{\gamma_1 - 1} x_{t-1} - \frac{\mu}{\gamma_1 - 1} \right) + \beta_0 \Delta x_t + \varepsilon_t$$

Interpretation for stationary variables: Suppose y_t and x_t are stationary with $Ey_t = \mu_y$ and $Ex_t = \mu_x$. These expectations are known as long run means. Then $E\Delta y_t = E\Delta x_t = 0$. We also have $E\varepsilon_t = 0$. Take expectations to see that

8 Cointegration (Dynamic Models)

The mission here is to test for cointegration in an ARDL model (as opposed to a simple AR model). There are two main methods.

Hendry's ARDL method:

1. Specify ARDL model, say,

$$y_t = \mu + \gamma_1 y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + \varepsilon$$

2. Find the ECM and long run equilibrium, say,

$$\Delta y_t = \alpha(y_{t-1} - \theta x_{t-1} - \tau) + \beta_0 \Delta x_t + \beta_1 x_{t-1} + \varepsilon_t$$

3. Estimation: use NLLS (Hendry) or OLS (Banerjee, Dolado, Mestre)

4. Test for no cointegration by testing $\alpha = 0$ or equivalently $\gamma_1 = 1$

5. Testing: Test for no cointegration by testing $H_0 : \alpha = 0$ vs $H_0 : \alpha < 0$. This is formulated as a $t_{\alpha=0}$ statistic for $\alpha = 0$. Critical values depend on properties of $y_t, x_t, \dots$. In particular, $\dim(x_t)$ is important in determining critical values (number of regressors not including the lags)

Remark. Consider for $\Delta y_t = \mu + (\gamma_1 - 1)y_{t-1} + (\beta_0 + \beta_1)x_{t-1} + \beta_0 \Delta x_t + \varepsilon_t$ which can be rewritten as

$$\Delta y_t = \alpha(y_{t-1} - \theta x_{t-1} + \beta_0 \Delta x_t + \varepsilon_t$$

The estimates for θ, τ are consistent (ratios of OLS estimators) and the standard distributions for θ, τ obtain when $-1 < \gamma_1 < 1$ or equivalently $-2 < \alpha < 0$. There could be more than one cointegrating vector and need to specify distribution of x_t since there could be feedback from y_t to x_t

Example 8.1. Consider the ARDL(1,1) model

$$y_t = \mu + \gamma_1 y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + \varepsilon_t$$

and the ECM

$$\Delta y_t = \alpha(y_{t-1} - \theta x_{t-1} - \tau) + \beta_0 \Delta x_t + \varepsilon_t$$

where $\alpha = \gamma_1 - 1$. We assume that (a) $y_t \sim I(1)$ and $x_t \sim I(1)$, (b) $-1 < \gamma \leq 1$ or equivalently $-2 < \alpha \leq 0$. In this model, y_t and x_t are cointegrated when $-2 < \alpha < 0$ and non-cointegrated when $\alpha = 0$.

We test for no cointegration by testing $H_0 : \alpha = 0$ vs $H_0 : \alpha < 0$.

We may estimate the ECM by OLS since

$$\begin{aligned} \Delta y_t &= \alpha(y_{t-1} - \theta x_{t-1} - \tau) + \beta_0 \Delta x_t + \varepsilon_t \\ &= \alpha y_{t-1} - \alpha \theta x_{t-1} + \alpha \tau + \beta_0 \Delta x_t + \varepsilon_t \\ &= \alpha y_{t-1} - \delta x_{t-1} - \mu + \beta_0 \Delta x_t + \varepsilon_t \end{aligned}$$

Test for no cointegration by testing $H_0 : \alpha = 0$ vs $H_0 : \alpha < 0$.

Johansen's Vector Autoregression Method:

1. Advantages:

- Allow for system with more variables, more cointegrating relations
- Better understanding of assumptions for unit root tests

The basic method of this approach is using VAR

$$\mathbf{X}_t = \Pi_1 \mathbf{X}_{t-1} + \Pi_2 \mathbf{X}_{t-2} + \cdots + \Pi_k \mathbf{X}_{t-k} + \Phi D_t + \varepsilon_t$$

- Uses likelihood methods for the analysis of cointegration in VAR models with Gaussian errors
- Likelihood ratio test of cointegration rank (nonstandard inference)
- Likelihood ratio test of cointegration rank (nonstandard inference)
- Tests of structural hypothesis about cointegrating relationships (standard inference)

Steps

1. Specify VAR model

$$\mathbf{X}_t = \Pi_1 \mathbf{X}_{t-1} + \Pi_2 \mathbf{X}_{t-2} + \cdots + \Pi_k \mathbf{X}_{t-k} + \Phi \mathbf{D}_t + \varepsilon_t$$

where $\mathbf{X}_t$ is a vector of p variables

$$\mathbf{X}_t = \begin{pmatrix} X_{1t} \\ X_{2t} \\ \vdots \\ X_{pt} \end{pmatrix}, \quad \mathbf{X}_{t-k} = \begin{pmatrix} X_{1t-k} \\ X_{2t-k} \\ \vdots \\ X_{pt-k} \end{pmatrix}$$

where D_t is a $p \times 1$ vector of deterministic components for $j = 1, \dots, k$, the matrices Π_j and the matrix Φ are $p \times p$ and $\varepsilon \sim \mathcal{N}_p(0, \Omega)$

2. Rewrite VAR to VECM: From our VAR model, rewrite

$$\Delta X_t = \Pi \mathbf{X}_{t-1} + \sum_{i=1}^{k-1} \Gamma_i \Delta \mathbf{X}_{t-i} + \Phi \mathbf{D}_t + \varepsilon_t$$

where $\Pi = \sum_{i=1}^k \Pi_i - I$ and $\Gamma_i = -\sum_{j=i+1}^k \Pi_j$

3. Testing: In the VECM model, we have three scenarios

- $\text{rank}(\Pi) = 0$, then $\Pi = 0 \implies \mathbf{X}_t \sim I(1)$ and No cointegration
- $\text{rank}(\Pi) = p$, then Π is full rank which implies $\mathbf{X}_t$ is stationary
- $\text{rank}(\Pi) = r \in [1, p-1]$, then $\Pi = \alpha\beta'$ reduced rank which implies $\mathbf{X}_t \sim I(1)$ and there is cointegration.

So the test for cointegration is essentially a test for rank. We have two such tests

- (a) **Trace test:** This procedure tests sequentially that there are at most r cointegrating vectors and hence $p - r$ unit roots. Hence, we test

$$\begin{aligned} H_0 : r = 0 \text{ v.s. } H_1 : r \leq p \\ H_0 : r \leq 1 \text{ v.s. } H_1 : r \leq p \\ H_0 : r \leq 2 \text{ v.s. } H_1 : r \leq p \\ \vdots \\ H_0 : r \leq p - 10 \text{ v.s. } H_1 : r \leq p \end{aligned}$$

I.e. we test $H_0 : \lambda_i = 0$ for $i = r + 1, \dots, p$ where λ_i is the i -th eigenvalue, with the test statistic/likelihood ratio statistic

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^p \ln(1 - \hat{\lambda}_i)$$

The sequence stops when it is not possible to reject a null hypothesis and there is a non-standard distribution under H_0 – simulated critical values available.

- (b) **Maximum eigenvalue test:** similarly a sequential test where we test that there are r cointegrating vectors against the alternative that are $r + 1$. The test statistic/likelihood ratio statistic in this case is

$$\lambda_{\text{max}}(r) = -T \ln(1 - \hat{\lambda}_{r+1}), \quad r = 0, 1, \dots, p - 1$$

for the hypotheses

$$\begin{aligned} H_0 : r = 0 \text{ v.s. } H_1 : r \leq 1 \\ H_0 : r \leq 1 \text{ v.s. } H_1 : r \leq 2 \\ H_0 : r \leq 2 \text{ v.s. } H_1 : r \leq 3 \\ \vdots \\ H_0 : r \leq p - 10 \text{ v.s. } H_1 : r \leq p \end{aligned}$$

with the sequence stopping once it is not possible to reject a null hypothesis. Nonstandard distribution under H_0 with simulated critical values.

Johansen approach in practice:

1. Specify and estimate a VAR(p) model for $\mathbf{Y}_t$
2. Test for cointegration (Johansen's test for reduced rank of Π)
3. If necessary, impose normalisation and identifying restrictions on the cointegrating vectors
4. Estimate the resulting cointegrated VECM by maximum likelihood
5. Test relevant hypothesis

The disadvantage is prima facie that we must model all variables but by modelling all variables, one gains understanding of assumptions for inference.

8.1 Full explication for representations of a cointegrated system

1. VAR
2. VECM
3. Common Trends
4. Moving Average
5. Triangular representation